

1. Introduction to free electron theory:

In the realm of solid-state physics, the behaviour of electrons in materials is a topic of great importance. Understanding how electrons move and interact in a solid is essential for explaining various electrical, thermal, and optical properties of materials. The free electron theory provides a simplified yet insightful model for comprehending the behaviour of electrons in metals. The electron theory has been developed in three main stages.

- (i) Classical free electron theory.
- (ii) Quantum free electron theory.
- (iii) Zone theory.

(i) **Classical free electron theory or (Drude and Lorentz theory of free electrons in metals)**

The Classical Free Electron Theory is a simplified model used in solid-state physics to describe the behaviour of electrons in metals. **Drude and Lorentz** introduced the initial theory in 1900. According to this theory, metals contain free electrons that account for electrical conductivity, and these electrons adhere to the principles of classical mechanics.

(ii) **Assumptions of Classical free electron theory (Drude and Lorentz theory of free electrons in metals):**

- (a) **Free Electron Assumption:** Electrons in a metal are assumed to be essentially free and not bound to specific atoms. They are considered to move freely throughout the entire metal lattice, experiencing the collective forces of the positively charged atomic cores.
- (b) **Neglect the Electron-Electron Interactions:** In this theory, electrons are treated as non-interacting particles, which simplifies the mathematical description of their behaviour. While this simplification is useful for some purposes, it does not account for electron-electron scattering events that can occur.
- (c) **Electron Gas Model:** This theory assumes the free electrons within the metal behave like gas molecules in the container and obeying the laws of kinetic theory of gases.
- (d) **Continuous Energy Levels:** Unlike electrons in isolated atoms, which have discrete energy levels, in the Classical Free Electron Theory, the energy levels of free electrons are treated as a continuum.
- (e) **Drift Velocity and Ohm's Law:** When an electric field is applied to a metal, the free electrons are assumed to respond to this field by drifting in the direction of the field. This drift of electrons constitutes an electric current, and the relationship between the applied electric field, drift velocity, and current density is described by Ohm's law.
- (f) Electron velocities in a metal obey Maxwell-Boltzmann distribution of velocities.

(g) In the absence of any external field the energy associated with each electron at temperature T is given by $E = \frac{3}{2} k_B T$, where k_B is Boltzmann constant.

(h) The electron move in a constant potential field.

(iii) **Advantages and disadvantages of classical free electron theory:**

Advantages:

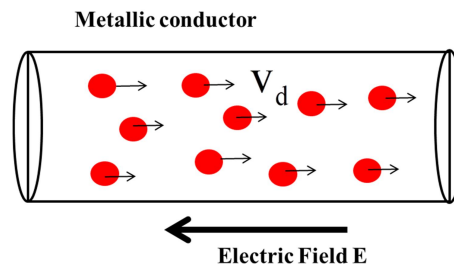
- i. It predicts the electrical conductivity and verifies the Ohm's law.
- ii. It gives the intuitive framework to understand the basic electrical and thermal properties of metals.
- iii. It explains optical properties of metals.
- iv. Derivation of the Widemann-Franz law.

Disadvantages:

- i. It fails to describe the behaviour of semiconductors, insulators, and other complex materials.
- ii. It fails to explain the changes in electrical conductivity with temperature, especially at low temperatures.
- iii. It does not provide an explanation for the specific heat of metals.
- iv. Classical free electron theory cannot account for phenomena such as the photoelectric effect, the Compton Effect, and black body radiation.
- v. It lacks to give an explanation for the temperature-dependent behaviours of paramagnetic susceptibility and ferromagnetism.

Definitions of quantities used in classical free electron theory:

1. **Drift velocity (v_d):** It is the average velocity attained by an electron under the influence by an electric field.



Suppose an electron of mass m , is set to move in an electric field E . Due to this electric field the electron will experience an electric force $F = eE$, opposite to the direction of electric field. Due to this force electron attains the velocity v_d

Now from Newton's second Law of motion:

$$F = ma = m \frac{d^2x}{dt^2} \dots \dots \dots (1)$$

Since we know that

$$F = eE \dots \dots \dots (2)$$

From equations (1) and (2), we can write

$$m \frac{d^2x}{dt^2} = eE$$

$$\Rightarrow \frac{d^2x}{dt^2} = \frac{eE}{m}$$

Integrating above equation, we have

$$\Rightarrow \frac{dx}{dt} = \frac{eE}{m} t + C$$

$$\text{Since at } t = 0 \Rightarrow \frac{dx}{dt} = 0$$

$$\Rightarrow C = 0$$

Therefore we have

$$v_d = \frac{dx}{dt} = \frac{eE}{m} \tau \dots \dots \dots (3)$$

Where $t = \tau$, is the relaxation time of the electron.

2. **Relaxation Time(τ):** Under the influence of external electric field, electrons attain the directional velocity v_d and if the electric field is switched off, velocity starts decreasing exponentially. Such a process which tends to restore the equilibrium is called *relaxation process*.

If $v_0 = v_d$ is the velocity at $t = 0$ at which the field is switch off. Now the velocity of the electron will starts decreasing exponentially.

$$v = v_d e^{-\frac{t}{\tau}}$$

Therefore the time taken by an electron to reduce its velocity $\frac{1}{e} v_d$ after the switch off of electric field is called relaxation Time.

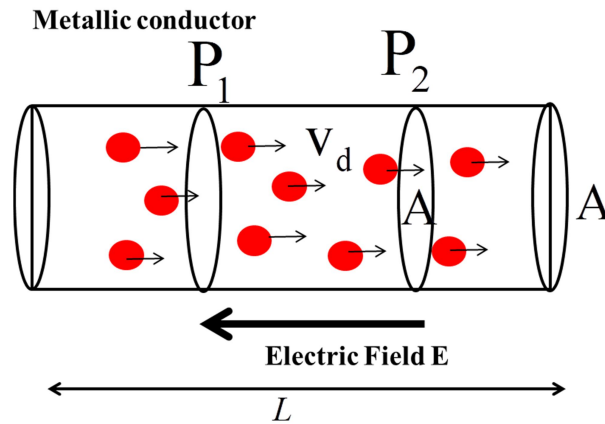
3. **Mean Free Path(λ):** The average distance travelled by the electrons between two successive collisions is known as mean free path.

4. **Mean Collision Time(τ_c):** Average time taken by the electron between two successive collisions.

$$\tau_c = \frac{\lambda}{v}$$

Where λ , is the mean free path and v is the velocity of the electron.

5. **Electrical conductivity in metals(σ):**



Let us consider a conductor of length L and cross-sectional area A .

So the Volume of the conductor = AL

Suppose the number of free electrons per unit volume = n

Therefore the total number of electrons in the conductor = nAL

Thus the total charge in the conductor $Q = enAL$

Suppose the time taken by the electrons to move from one end to another.

Therefore the current i in the conductor,

$$i = \frac{Q}{t} = \frac{enAL}{t} \dots \dots \dots (1)$$

Since the electrons moves in the conductor with drift velocity v_d ,

$$\text{Therefore } v_d = \frac{L}{t} \dots \dots \dots (2)$$

Thus the equation (1) can be written as

$$i = n e A v_d \dots \dots \dots (3)$$

Current Density (J): It is defined as the current per unit of area.

$$J = \frac{i}{A} = n e v_d \dots \dots \dots (4)$$

Again from the microscopic form of Ohm's Law

$$J = \sigma E \dots \dots \dots (5)$$

Where, σ , is the electrical conductivity of the conductor, and E , is applied electric field.

From equations (4) and (5), we can write

$$\sigma = \frac{J}{E} = \frac{n e v_d}{E} \dots \dots \dots (6)$$

Since we know that $v_d = \frac{dx}{dt} = \frac{eE}{m} \tau \dots \dots \dots (7)$

Therefore we can write

$$\sigma = \frac{J}{E} = \frac{n e v_d}{E} = \frac{n e^2 \tau}{m} \dots \dots \dots (8)$$

Unit of electrical conductivity is ***ohm⁻¹ – meter⁻¹ or siemens/meter***

Resistivity(ρ): Resistivity of a conductor is the reciprocal of the electrical conductivity.

$$\rho = \frac{1}{\sigma} = \frac{m}{n e^2 \tau} \dots \dots \dots (9)$$

Unit of resistivity is ***ohm-meter***

Mobility of the electrons: It is defined as drift velocity per unit electric field.

$$\mu = \frac{v_d}{E} = \frac{\sigma}{n e} \dots \dots \dots (10)$$

$$\Rightarrow \sigma = n e \mu \dots \dots \dots (11)$$

Unit of mobility *meter²/Volt.Sec*

6. Temperature dependence of electrical conductivity:

Since we know that electrical conductivity of the metal is given by (as discussed in equation (8))

$$\sigma = \frac{n e^2 \tau}{m} \dots \dots \dots (12)$$

Further we know that Average energy of an electron is given by

$$E_{ave} = \frac{3}{2} k_B T \dots \dots \dots (13)$$

This average energy in the electron, will be in the form of kinetic energy,

$$E_{ave} = \text{K.E.} = \frac{mv_{th}^2}{2} \dots\dots\dots(14)$$

Where v_{th} , is the thermal velocity of the electron at temperature T, and it is given by

$$v_{th} = \sqrt{\frac{3k_B T}{m}} \dots\dots\dots(15)$$

Now the collision time is inversely proportional to thermal velocity

$$\tau \propto \frac{1}{v_{th}}$$

$$\Rightarrow \tau \propto \frac{1}{\sqrt{T}}$$

Therefore from equation (1) (Theoretical prediction)

$$\sigma \propto \frac{1}{\sqrt{T}} \dots\dots\dots(16)$$

But experimental observation says that

$$\sigma \propto \frac{1}{T} \dots\dots\dots(17)$$

Hence from equations (16) & (17), prediction of classical free electron theory is not in agreement with the experimental observations. Thus the classical free electron theory fails to explain dependence of T on σ .